Database Management Systems

Lab-09

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Section:G

Output Screenshots:

Task 1:

Q1:

```
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]> -- Q1: Resident payment leaders
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]>
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]> WITH PaymentTotals AS (
    ->     SELECT
    ->         Student_ID,
    ->         SUM(Amount) AS Resident_Total
    ->     FROM PAYMENT
    ->     GROUP BY Student_ID
    -> ),
    -> CurrentResidents AS (
    ->     SELECT
    ->         A.Hostel_ID,
    ->         A.Student_ID,
    ->         CONCAT(S.First_Name, ' ', S.Last_Name) AS Student_Name
    ->     FROM ALLOCATE A
    ->     JOIN STUDENT S
    ->         ON S.Student_ID = A.Student_ID
    ->     WHERE A.Check_in_Date IS NOT NULL
    ->       AND A.Check_out_Date IS NULL
    -> ),
    -> ResidentTotals AS (
    ->     SELECT
    ->         CR.Hostel_ID,
    ->         CR.Student_ID,
    ->         CR.Student_Name,
    ->         COALESCE(PT.Resident_Total, 0) AS Resident_Total
    ->     FROM CurrentResidents CR
    ->     LEFT JOIN PaymentTotals PT
    ->         ON PT.Student_ID = CR.Student_ID
    -> ),
    -> HostelAverages AS (
    ->     SELECT
    ->         H.Hostel_ID,
    ->         H.Hostel_Name,
    ->         AVG(RT.Resident_Total) AS Hostel_Avg
    ->     FROM HOSTEL H
    ->     JOIN ResidentTotals RT
    ->         ON RT.Hostel_ID = H.Hostel_ID
    ->     GROUP BY H.Hostel_ID, H.Hostel_Name
    -> ),
```

```
  -> CampusBenchmark AS (
  ->     SELECT AVG(Resident_Total) AS Benchmark
  ->     FROM ResidentTotals
  -> ),
  -> QualifyingHostels AS (
  ->     SELECT
  ->         HA.Hostel_ID,
  ->         HA.Hostel_Name,
  ->         HA.Hostel_Avg,
  ->         CB.Benchmark
  ->     FROM HostelAverages HA
  ->     CROSS JOIN CampusBenchmark CB
  ->     WHERE HA.Hostel_Avg > CB.Benchmark
  -> ),
  -> HostelMaximums AS (
  ->     SELECT
  ->         RT.Hostel_ID,
  ->         MAX(RT.Resident_Total) AS Max_Total
  ->     FROM ResidentTotals RT
  ->     JOIN QualifyingHostels QH
  ->         ON QH.Hostel_ID = RT.Hostel_ID
  ->     GROUP BY RT.Hostel_ID
  -> )
  -> SELECT
  ->     QH.Hostel_ID,
  ->     QH.Hostel_Name,
  ->     ROUND(QH.Hostel_Avg, 2) AS Hostel_Avg,
  ->     ROUND(QH.Benchmark, 2) AS Benchmark,
  ->     RT.Student_ID,
  ->     RT.Student_Name,
  ->     ROUND(RT.Resident_Total, 2) AS Resident_Total
  -> FROM QualifyingHostels QH
  -> JOIN HostelMaximums HM
  ->     ON HM.Hostel_ID = QH.Hostel_ID
  -> JOIN ResidentTotals RT
  ->     ON RT.Hostel_ID = HM.Hostel_ID
  ->    AND RT.Resident_Total = HM.Max_Total
  -> ORDER BY QH.Hostel_ID, RT.Student_ID;
+-----------+-------------+------------+-----------+------------+--------------+----------------+
| Hostel_ID | Hostel_Name | Hostel_Avg | Benchmark | Student_ID | Student_Name | Resident_Total |
+-----------+-------------+------------+-----------+------------+--------------+----------------+
| H003      | C Block     |   39727.27 |  36467.92 | S021       | Varun Kumar  |       57000.00 |
| H003      | C Block     |   39727.27 |  36467.92 | S026       | Shreya Iyer  |       57000.00 |
| H005      | E Block     |   38000.00 |  36467.92 | S041       | Diya Rao     |       73000.00 |
+-----------+-------------+------------+-----------+------------+--------------+----------------+
```

Q2:

```
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]> WITH RECURSIVE StaffAncestors AS (
    ->     SELECT
    ->         S.Staff_ID,
    ->         S.Supervisor_ID AS Ancestor_ID,
    ->         CAST(
    ->             CONCAT('/', S.Staff_ID, '/', S.Supervisor_ID, '/')
    ->             AS CHAR(2000)
    ->         ) AS Path
    ->     FROM STAFF S
    ->     WHERE S.Supervisor_ID IS NOT NULL
    ->
    ->     UNION ALL
    ->
    ->     SELECT
    ->         SA.Staff_ID,
    ->         Supervisor.Supervisor_ID AS Ancestor_ID,
    ->         CONCAT(SA.Path, Supervisor.Supervisor_ID, '/') AS Path
    ->     FROM StaffAncestors SA
    ->     JOIN STAFF Supervisor
    ->         ON Supervisor.Staff_ID = SA.Ancestor_ID
    ->     WHERE Supervisor.Supervisor_ID IS NOT NULL
    ->       AND LOCATE(
    ->           CONCAT('/', Supervisor.Supervisor_ID, '/'),
    ->           SA.Path
    ->       ) = 0
    -> ),
    -> InversionSummary AS (
    ->     SELECT
    ->         S.Staff_ID,
    ->         S.Name,
    ->         S.Salary,
    ->         SUM(
    ->             CASE
    ->                 WHEN S.Salary > A.Salary THEN 1
    ->                 ELSE 0
    ->             END
    ->         ) AS Inversion_Count,
    ->         MAX(
    ->             CASE
    ->                 WHEN S.Salary > A.Salary
    ->                 THEN S.Salary - A.Salary
    ->                 ELSE NULL
    ->             END
    ->         ) AS Max_Gap
```

```
    ->         S.Salary,
    ->         SUM(
    ->             CASE
    ->                 WHEN S.Salary > A.Salary THEN 1
    ->                 ELSE 0
    ->             END
    ->         ) AS Inversion_Count,
    ->         MAX(
    ->             CASE
    ->                 WHEN S.Salary > A.Salary
    ->                 THEN S.Salary - A.Salary
    ->                 ELSE NULL
    ->             END
    ->         ) AS Max_Gap
    ->     FROM STAFF S
    ->     JOIN StaffAncestors SA
    ->         ON SA.Staff_ID = S.Staff_ID
    ->     JOIN STAFF A
    ->         ON A.Staff_ID = SA.Ancestor_ID
    ->     GROUP BY
    ->         S.Staff_ID,
    ->         S.Name,
    ->         S.Salary
    -> )
    -> SELECT
    ->     Staff_ID,
    ->     Name,
    ->     Salary,
    ->     Inversion_Count,
    ->     ROUND(Max_Gap, 2) AS Max_Gap
    -> FROM InversionSummary
    -> WHERE Inversion_Count > 0
    -> ORDER BY
    ->     Inversion_Count DESC,
    ->     Staff_ID;
+----------+---------------+----------+-----------------+---------+
| Staff_ID | Name          | Salary   | Inversion_Count | Max_Gap |
+----------+---------------+----------+-----------------+---------+
| ST032    | Neeraj Pillai | 40000.00 |               3 | 4000.00 |
| ST028    | Prakash Shetty | 39000.00 |               1 | 3000.00 |
| ST029    | Imran Khan    | 38000.00 |               1 | 2000.00 |
| ST030    | Farah Siddiqui | 36000.00 |               1 | 1000.00 |
| ST033    | Lalitha Murthy | 46000.00 |               1 | 1000.00 |
+----------+---------------+----------+-----------------+---------+
5 rows in set (0.00 sec)
```

Q3:

```
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]> -- Q3: Room Occupancy Reconciliation
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]>
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]> WITH BedCounts AS (
    ->     SELECT
    ->         Hostel_ID,
    ->         Room_Number,
    ->         COUNT(*) AS Total_Beds,
    ->         SUM(
    ->             CASE
    ->                 WHEN Availability = 'N' THEN 1
    ->                 ELSE 0
    ->             END
    ->         ) AS Flags
    ->     FROM BED
    ->     GROUP BY
    ->         Hostel_ID,
    ->         Room_Number
    -> ),
    -> ResidentCounts AS (
    ->     SELECT
    ->         Hostel_ID,
    ->         Room_Number,
    ->         COUNT(*) AS Residents
    ->     FROM ALLOCATE
    ->     WHERE Check_in_Date IS NOT NULL
    ->       AND Check_out_Date IS NULL
    ->     GROUP BY
    ->         Hostel_ID,
    ->         Room_Number
    -> ),
    -> RoomReconciliation AS (
    ->     SELECT
    ->         R.Hostel_ID,
    ->         R.Room_Number,
    ->         COALESCE(B.Total_Beds, 0) AS Total_Beds,
    ->         COALESCE(B.Flags, 0) AS Flags,
    ->         COALESCE(RC.Residents, 0) AS Residents,
    ->         CASE
    ->             WHEN COALESCE(B.Total_Beds, 0) = 0
    ->                 THEN 'NO_BEDS'
    ->             WHEN COALESCE(B.Flags, 0) = COALESCE(RC.Residents, 0)
    ->                 THEN 'MATCH'
    ->             WHEN COALESCE(B.Flags, 0) > COALESCE(RC.Residents, 0)
    ->                 THEN 'FLAGS_EXCEED'
```

```
    ->     LEFT JOIN BedCounts B
    ->         ON B.Hostel_ID = R.Hostel_ID
    ->        AND B.Room_Number = R.Room_Number
    ->     LEFT JOIN ResidentCounts RC
    ->         ON RC.Hostel_ID = R.Hostel_ID
    ->        AND RC.Room_Number = R.Room_Number
    -> )
    -> SELECT
    ->     H.Hostel_ID,
    ->     H.Hostel_Name,
    ->     SUM(
    ->         CASE WHEN RR.Reconciliation_Status = 'NO_BEDS'
    ->              THEN 1 ELSE 0 END
    ->     ) AS No_Beds_Rooms,
    ->     SUM(
    ->         CASE WHEN RR.Reconciliation_Status = 'MATCH'
    ->              THEN 1 ELSE 0 END
    ->     ) AS Match_Rooms,
    ->     SUM(
    ->         CASE WHEN RR.Reconciliation_Status = 'FLAGS_EXCEED'
    ->              THEN 1 ELSE 0 END
    ->     ) AS Flags_Exceed_Rooms,
    ->     SUM(
    ->         CASE WHEN RR.Reconciliation_Status = 'RESIDENTS_EXCEED'
    ->              THEN 1 ELSE 0 END
    ->     ) AS Residents_Exceed_Rooms,
    ->     SUM(RR.Correction) AS Net_Correction
    -> FROM HOSTEL H
    -> LEFT JOIN RoomReconciliation RR
    ->     ON RR.Hostel_ID = H.Hostel_ID
    -> GROUP BY
    ->     H.Hostel_ID,
    ->     H.Hostel_Name
    -> ORDER BY H.Hostel_ID;
+-----------+-------------+---------------+-------------+--------------------+------------------------+----------------+
| Hostel_ID | Hostel_Name | No_Beds_Rooms | Match_Rooms | Flags_Exceed_Rooms | Residents_Exceed_Rooms | Net_Correction |
+-----------+-------------+---------------+-------------+--------------------+------------------------+----------------+
| H001      | A Block     |             0 |           1 |                  3 |                      2 |             -2 |
| H002      | B Block     |             1 |           3 |                  2 |                      1 |             -2 |
| H003      | C Block     |             0 |           5 |                  0 |                      1 |              1 |
| H004      | D Block     |             0 |           1 |                  3 |                      2 |             -2 |
| H005      | E Block     |             0 |           3 |                  1 |                      2 |              1 |
+-----------+-------------+---------------+-------------+--------------------+------------------------+----------------+
5 rows in set (0.00 sec)
```

Q4:

```
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]> -- Q4: Staff Salary Revision
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]>
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]> WITH RECURSIVE RaisedSalaries AS (
    ->     SELECT
    ->         Staff_ID,
    ->         Name,
    ->         Salary,
    ->         Hostel_ID,
    ->         Supervisor_ID,
    ->         CASE
    ->             WHEN Salary < 25000
    ->                 THEN ROUND(Salary * 1.12, 2)
    ->             WHEN Salary < 35000
    ->                 THEN ROUND(Salary * 1.08, 2)
    ->             ELSE ROUND(Salary * 1.05, 2)
    ->         END AS Raised_Salary
    ->     FROM STAFF
    -> ),
    -> RevisedStaff AS (
    ->     SELECT
    ->         RS.Staff_ID,
    ->         RS.Name,
    ->         RS.Salary,
    ->         RS.Hostel_ID,
    ->         RS.Supervisor_ID,
    ->         RS.Raised_Salary,
    ->         RS.Raised_Salary AS Revised_Salary
    ->     FROM RaisedSalaries RS
    ->     WHERE RS.Supervisor_ID IS NULL
    ->
    ->     UNION ALL
    ->
    ->     SELECT
    ->         Child.Staff_ID,
    ->         Child.Name,
    ->         Child.Salary,
    ->         Child.Hostel_ID,
    ->         Child.Supervisor_ID,
    ->         Child.Raised_Salary,
    ->         ROUND(
    ->             LEAST(
    ->                 Child.Raised_Salary,
```

```
    ->                 Parent.Revised_Salary
    ->             ),
    ->             2
    ->         ) AS Revised_Salary
    ->     FROM RaisedSalaries Child
    ->     JOIN RevisedStaff Parent
    ->         ON Child.Supervisor_ID = Parent.Staff_ID
    -> ),
    -> HostelPayroll AS (
    ->     SELECT
    ->         H.Hostel_ID,
    ->         H.Hostel_Name,
    ->         SUM(RS.Salary) AS Current_Payroll,
    ->         SUM(RS.Revised_Salary) AS Revised_Payroll
    ->     FROM HOSTEL H
    ->     LEFT JOIN RevisedStaff RS
    ->         ON RS.Hostel_ID = H.Hostel_ID
    ->     GROUP BY
    ->         H.Hostel_ID,
    ->         H.Hostel_Name
    -> )
    -> SELECT
    ->     Hostel_ID,
    ->     Hostel_Name,
    ->     ROUND(COALESCE(Current_Payroll, 0), 2) AS Current_Payroll,
    ->     ROUND(COALESCE(Revised_Payroll, 0), 2) AS Revised_Payroll,
    ->     ROUND(
    ->         COALESCE(Revised_Payroll, 0)
    ->         - COALESCE(Current_Payroll, 0),
    ->         2
    ->     ) AS Increase
    -> FROM HostelPayroll
    -> ORDER BY Hostel_ID;
+-----------+-------------+-----------------+-----------------+----------+
| Hostel_ID | Hostel_Name | Current_Payroll | Revised_Payroll | Increase |
+-----------+-------------+-----------------+-----------------+----------+
| H001      | A Block     |       250000.00 |       267060.00 | 17060.00 |
| H002      | B Block     |       166000.00 |       177750.00 | 11750.00 |
| H003      | C Block     |       151500.00 |       163280.00 | 11780.00 |
| H004      | D Block     |       288500.00 |       296960.00 |  8460.00 |
| H005      | E Block     |       216000.00 |       229780.00 | 13780.00 |
+-----------+-------------+-----------------+-----------------+----------+
5 rows in set (0.01 sec)
```

Q5:

```
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]> -- Q5: Facility Engagement Profile
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]>
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]> WITH CurrentResidents AS (
    ->     SELECT
    ->         Hostel_ID,
    ->         Student_ID
    ->     FROM ALLOCATE
    ->     WHERE Check_in_Date IS NOT NULL
    ->       AND Check_out_Date IS NULL
    -> ),
    -> ResidentEngagement AS (
    ->     SELECT
    ->         CR.Hostel_ID,
    ->         CR.Student_ID,
    ->         COUNT(DISTINCT F.Facility_ID) AS Home_Facility_Count
    ->     FROM CurrentResidents CR
    ->     LEFT JOIN USES U
    ->         ON U.Student_ID = CR.Student_ID
    ->     LEFT JOIN FACILITY F
    ->         ON F.Facility_ID = U.Facility_ID
    ->        AND F.Hostel_ID = CR.Hostel_ID
    ->     GROUP BY
    ->         CR.Hostel_ID,
    ->         CR.Student_ID
    -> ),
    -> ClassifiedResidents AS (
    ->     SELECT
    ->         Hostel_ID,
    ->         Student_ID,
    ->         CASE
    ->             WHEN Home_Facility_Count = 0 THEN 'Inactive'
    ->             WHEN Home_Facility_Count = 1 THEN 'Low'
    ->             WHEN Home_Facility_Count = 2 THEN 'Medium'
    ->             ELSE 'High'
    ->         END AS Engagement_Class
    ->     FROM ResidentEngagement
    -> ),
```

```
    ->         Student_ID,
    ->         CASE
    ->             WHEN Home_Facility_Count = 0 THEN 'Inactive'
    ->             WHEN Home_Facility_Count = 1 THEN 'Low'
    ->             WHEN Home_Facility_Count = 2 THEN 'Medium'
    ->             ELSE 'High'
    ->         END AS Engagement_Class
    ->     FROM ResidentEngagement
    -> ),
    -> ClassCounts AS (
    ->     SELECT
    ->         Hostel_ID,
    ->         Engagement_Class,
    ->         COUNT(*) AS Resident_Count
    ->     FROM ClassifiedResidents
    ->     GROUP BY
    ->         Hostel_ID,
    ->         Engagement_Class
    -> )
    -> SELECT
    ->     CC.Hostel_ID,
    ->     CC.Engagement_Class,
    ->     CC.Resident_Count
    -> FROM ClassCounts CC
    -> WHERE CC.Resident_Count = (
    ->     SELECT MAX(CC2.Resident_Count)
    ->     FROM ClassCounts CC2
    ->     WHERE CC2.Hostel_ID = CC.Hostel_ID
    -> )
    -> ORDER BY
    ->     CC.Hostel_ID,
    ->     CC.Engagement_Class;
+-----------+------------------+----------------+
| Hostel_ID | Engagement_Class | Resident_Count |
+-----------+------------------+----------------+
| H001      | Medium           |              6 |
| H002      | Medium           |              5 |
| H003      | Medium           |              6 |
| H004      | Medium           |              4 |
| H005      | Low              |              4 |
| H005      | Medium           |              4 |
+-----------+------------------+----------------+
6 rows in set (0.00 sec)

PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]>
```

Q6:

```
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]> -- Q6: Daily Fee Ledger
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]>
PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]> WITH RECURSIVE
    -> CurrentResidents AS (
    ->     SELECT
    ->         Student_ID,
    ->         Hostel_ID,
    ->         Room_Number,
    ->         Check_in_Date
    ->     FROM ALLOCATE
    ->     WHERE Check_in_Date IS NOT NULL
    ->       AND Check_out_Date IS NULL
    ->       AND Check_in_Date <= '2025-08-20'
    -> ),
    -> PaymentByDay AS (
    ->     SELECT
    ->         Student_ID,
    ->         Payment_Date,
    ->         SUM(Amount) AS Payment_Amount
    ->     FROM PAYMENT
    ->     WHERE Payment_Date <= '2025-08-20'
    ->     GROUP BY
    ->         Student_ID,
    ->         Payment_Date
    -> ),
    -> InitialPayments AS (
    ->     SELECT
    ->         CR.Student_ID,
    ->         COALESCE(SUM(P.Amount), 0) AS Total_Payments
    ->     FROM CurrentResidents CR
    ->     LEFT JOIN PAYMENT P
    ->         ON P.Student_ID = CR.Student_ID
    ->        AND P.Payment_Date <= CR.Check_in_Date
    ->     GROUP BY CR.Student_ID
    -> ),
    -> Ledger (
    ->     Student_ID,
    ->     Hostel_ID,
    ->     Room_Number,
    ->     Check_in_Date,
    ->     Ledger_Date,
    ->     Days_Charged,
    ->     Balance,
    ->     Total_Late_Fees,
    ->     Total_Payments_Applied
    -> ) AS (
    ->     SELECT
    ->         CR.Student_ID,
    ->         CR.Hostel_ID,
```

```
    ->     Total_Payments_Applied
    -> ) AS (
    ->     SELECT
    ->         CR.Student_ID,
    ->         CR.Hostel_ID,
    ->         CR.Room_Number,
    ->         CR.Check_in_Date,
    ->         CR.Check_in_Date AS Ledger_Date,
    ->         1 AS Days_Charged,
    ->         CAST(
    ->             (100 - IP.Total_Payments)
    ->             + CASE
    ->                 WHEN 100 - IP.Total_Payments > 0
    ->                     THEN ROUND(
    ->                         (100 - IP.Total_Payments) * 0.02,
    ->                         2
    ->                     )
    ->                 ELSE 0
    ->               END
    ->             AS DECIMAL(14,2)
    ->         ) AS Balance,
    ->         CAST(
    ->             CASE
    ->                 WHEN 100 - IP.Total_Payments > 0
    ->                     THEN ROUND(
    ->                         (100 - IP.Total_Payments) * 0.02,
    ->                         2
    ->                     )
    ->                 ELSE 0
    ->             END
    ->             AS DECIMAL(14,2)
    ->         ) AS Total_Late_Fees,
    ->         CAST(
    ->             IP.Total_Payments
    ->             AS DECIMAL(14,2)
    ->         ) AS Total_Payments_Applied
    ->     FROM CurrentResidents CR
    ->     JOIN InitialPayments IP
    ->         ON IP.Student_ID = CR.Student_ID
    ->
    ->     UNION ALL
    ->
    ->     SELECT
    ->         L.Student_ID,
    ->         L.Hostel_ID,
    ->         L.Room_Number,
    ->         L.Check_in_Date,
    ->         DATE_ADD(L.Ledger_Date, INTERVAL 1 DAY),
    ->         L.Days_Charged + 1,
    ->         CAST(
    ->             (L.Balance + 100 - COALESCE(P.Payment_Amount, 0))
    ->             + CASE
    ->                 WHEN L.Balance + 100
    ->                      - COALESCE(P.Payment_Amount, 0) > 0
    ->                     THEN ROUND(
    ->                         (L.Balance + 100
```

```
    -> ORDER BY Student_ID;
+------------+-----------+-------------+--------------+-----------------+-----------------+-----------------------+---------------+
| Student_ID | Hostel_ID | Room_Number | Days_Charged | Total_Daily_Fees | Total_Late_Fees | Total_Payments_Applied | Final_Balance |
+------------+-----------+-------------+--------------+-----------------+-----------------+-----------------------+---------------+
| S001       | H001      | 101         |           47 |            4700 |           30.81 |              80000.00 |      -75269.19 |
| S002       | H001      | 101         |           47 |            4700 |           43.43 |              28000.00 |      -23256.57 |
| S003       | H001      | 101         |           47 |            4700 |           58.30 |              22000.00 |      -17241.70 |
| S004       | H001      | 102         |           47 |            4700 |           58.30 |              30000.00 |      -25241.70 |
| S005       | H001      | 102         |           47 |            4700 |           75.47 |              39000.00 |      -34224.53 |
| S006       | H001      | 102         |           47 |            4700 |           75.47 |              24000.00 |      -19224.53 |
| S007       | H001      | 103         |           47 |            4700 |           94.98 |              32000.00 |      -27205.02 |
| S008       | H001      | 103         |           47 |            4700 |           94.98 |              26000.00 |      -21205.02 |
| S009       | H001      | 103         |           47 |            4700 |          116.88 |              21000.00 |      -16183.12 |
| S010       | H001      | 201         |           47 |            4700 |          116.88 |              35000.00 |      -30183.12 |
| S011       | H002      | 101         |           46 |            4600 |          116.88 |              78000.00 |      -73283.12 |
| S012       | H002      | 101         |           46 |            4600 |          116.88 |              18000.00 |      -13283.12 |
| S013       | H002      | 101         |           46 |            4600 |          141.22 |              29000.00 |      -24258.78 |
| S014       | H002      | 102         |           46 |            4600 |          141.22 |              23000.00 |      -18258.78 |
| S015       | H002      | 102         |           46 |            4600 |          168.04 |              60000.00 |      -55231.96 |
| S016       | H002      | 102         |           46 |            4600 |          168.04 |              20000.00 |      -15231.96 |
| S017       | H002      | 103         |           46 |            4600 |          197.40 |              31000.00 |      -26202.60 |
| S018       | H002      | 103         |           46 |            4600 |          197.40 |              27000.00 |      -22202.60 |
| S019       | H002      | 103         |           46 |            4600 |          229.35 |              38000.00 |      -33170.65 |
| S020       | H002      | 104         |           46 |            4600 |          229.35 |              24000.00 |      -19170.65 |
| S021       | H003      | 101         |           45 |            4500 |          229.35 |              57000.00 |      -52270.65 |
| S022       | H003      | 101         |           45 |            4500 |          229.35 |              42000.00 |      -37270.65 |
| S023       | H003      | 102         |           45 |            4500 |          263.94 |              26000.00 |      -21236.06 |
| S024       | H003      | 102         |           45 |            4500 |          263.94 |              33000.00 |      -28236.06 |
| S025       | H003      | 201         |           45 |            4500 |          301.22 |              37000.00 |      -32198.78 |
| S026       | H003      | 201         |           45 |            4500 |          301.22 |              57000.00 |      -52198.78 |
| S027       | H003      | 202         |           45 |            4500 |          341.24 |              28000.00 |      -23158.76 |
| S028       | H003      | 202         |           45 |            4500 |          341.24 |              36000.00 |      -31158.76 |
| S029       | H003      | 301         |           45 |            4500 |          384.06 |              22000.00 |      -17115.94 |
| S030       | H003      | 301         |           45 |            4500 |          384.06 |              50000.00 |      -45115.94 |
| S031       | H004      | 101         |           44 |            4400 |          384.06 |              82000.00 |      -77215.94 |
| S032       | H004      | 101         |           44 |            4400 |          384.06 |              17000.00 |      -12215.94 |
| S033       | H004      | 101         |           44 |            4400 |          429.74 |              44000.00 |      -39170.26 |
| S034       | H004      | 102         |           44 |            4400 |          429.74 |              25000.00 |      -20170.26 |
| S035       | H004      | 102         |           44 |            4400 |          478.33 |              43000.00 |      -38121.67 |
| S036       | H004      | 102         |           44 |            4400 |          478.33 |              54800.00 |      -49921.67 |
| S037       | H004      | 103         |           44 |            4400 |          341.24 |              26000.00 |      -21258.76 |
| S038       | H004      | 103         |           44 |            4400 |          529.90 |              46000.00 |      -41070.10 |
| S039       | H004      | 103         |           44 |            4400 |          584.50 |              27000.00 |      -22015.50 |
| S040       | H004      | 201         |           44 |            4400 |          584.50 |              36000.00 |      -31015.50 |
| S041       | H005      | 101         |           43 |            4300 |          584.50 |              73000.00 |      -68115.50 |
| S042       | H005      | 101         |           43 |            4300 |          642.19 |              41000.00 |      -36057.81 |
| S043       | H005      | 102         |           43 |            4300 |          703.03 |              29000.00 |      -23996.97 |
| S044       | H005      | 102         |           43 |            4300 |          703.03 |              38000.00 |      -32996.97 |
| S045       | H005      | 201         |           43 |            4300 |          767.09 |              27000.00 |      -21932.91 |
| S046       | H005      | 201         |           43 |            4300 |          767.09 |              48000.00 |      -42932.91 |
| S047       | H005      | 202         |           43 |            4300 |          834.43 |              26000.00 |      -20865.57 |
| S048       | H005      | 202         |           43 |            4300 |          834.43 |              43000.00 |      -37865.57 |
| S049       | H005      | 301         |           43 |            4300 |          905.12 |              20000.00 |      -14794.88 |
| S050       | H005      | 301         |           43 |            4300 |          905.12 |              35000.00 |      -29794.88 |
| S052       | H003      | 302         |           19 |            1900 |            2.00 |              49000.00 |      -47098.00 |
| S055       | H004      | 202         |            6 |             600 |           43.43 |                  0.00 |         643.43 |
| S056       | H002      | 203         |           15 |            1500 |            2.00 |              30000.00 |      -28498.00 |
+------------+-----------+-------------+--------------+-----------------+-----------------+-----------------------+---------------+
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PES2UG24CS387_YOURNAME@localhost [HOSTEL_LAB9]>
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